PURDUE UNIVERSITY | Elmore Family School of Electrical and Computer Engineering

ECE 49022 – Electrical Engineering Senior Design Projects
Fall 2026 – Product-Level Engineering Design Requirements Report

MASHD: Multifunction Appliance for Smart Home Display

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1.0 Project Product Subsystems List:

1. User Display
2. Processing Subsystem
3. Wireless Subsystem
4. Sensor Suite
5. Power System
6. Mechanical System

2.0 Project-Level Engineering Design Requirement Statements for your Base Product:

1) The product shall be clearly visible from across the room to someone with 20-20 vision or corrective lenses.
2) The product display shall be settable by the user from their mobile device.
3) The product should have a low power mode setting in the case of no person being present.
4) The product shall be wirelessly connected to a mobile phone.
5) The product shall monitor human activity, noise, light, temperature, and humidity.
6) The product shall be powered over a cable coming from a wall outlet.
7) The product shall supply the necessary voltages at the necessary currents for every component.
8) The product shall be wall mountable.
9) The product shall have appropriate mounts and holes for all the internal electrical components.
10) The product shall maintain temperatures below maximum operating conditions of 60ºC.
11) The product should pull weather data, calendar information, live flight data, and other internet-enabled features from the internet.

3.0 Stretch Project-Level Engineering Design Requirement Statements:

S1) The product could flash firmware updates over Wi-Fi to all compute modules.
S2) The product could set images beyond strings to the display.
S3) There product could have an additional admin interface present that is password protected